A sum of cube roots is irrational

Cambridge Tripos, 2022 IA 4-I-2D (a)

Show that $\sqrt[3]{2} + \sqrt[3]{3}$ is irrational.

Proof. Put

$$a = 2^{1/3}, \qquad b = 3^{1/3}, \qquad w = 6^{1/3},$$

where the powers are real powers. For $x \geq 0$, the laws of real powers give

$$\left(x^{1/3}\right)^3 = x^{(1/3)\cdot 3} = x^1 = x.$$

In particular,

$$a^3 = 2, \qquad b^3 = 3, \qquad w^3 = 6,$$

and a, b, w are all positive, since $2, 3, 6 > 0$.

Moreover, by multiplicativity of real powers for nonnegative bases,

$$ab = 2^{1/3} \cdot 3^{1/3} = (2 \cdot 3)^{1/3} = 6^{1/3} = w.$$

Now expand the cube of $a + b$:

$$(a+b)^3 = a^3 + b^3 + 3ab(a+b).$$

Using $a^3 = 2$, $b^3 = 3$, and $ab = w$, we obtain

$$(a+b)^3 = 5 + 3w(a+b). \tag{1}$$

We first show that w is irrational. Suppose instead that w is rational, say $w = q$ for some $q \in \mathbb{Q}$. Cubing gives $q^3 = 6$ in $\mathbb{Q}$. Write q in lowest terms: let $u = |\text{numerator of } q| \in \mathbb{N}$ and $v = \text{denominator of } q \in \mathbb{N}$, so that $\gcd(u, v) = 1$, and taking absolute values in $q^3 = 6$ yields

$$u^3 = 6v^3. \tag{2}$$

We use the following fact: if $x \in \mathbb{N}$ and $2 \mid x^3$, then $2 \mid x$. Indeed, $x^3 = x \cdot x \cdot x$, and since 2 is prime, 2 dividing this product forces 2 to divide one of the factors, i.e. $2 \mid x$.

From (2), $u^3 = 2 \cdot (3v^3)$, so $2 \mid u^3$ and hence $2 \mid u$. Write $u = 2u_1$ with $u_1 \in \mathbb{N}$. Substituting into (2) gives

$$8u_1^3 = 6v^3, \qquad \text{hence} \qquad 4u_1^3 = 3v^3.$$

Thus $2 \mid 3v^3$. Since 2 is prime and $2 \nmid 3$, we get $2 \mid v^3$, and hence $2 \mid v$ by the same fact. Therefore 2 divides both u and v, so $2 \mid \gcd(u, v) = 1$, a contradiction. Hence w is irrational.

Now set $s = a + b$. Then $s > 0$, in particular $s \neq 0$, so we may rearrange (1) to get

$$w = \frac{s^3 - 5}{3s}. \tag{3}$$

Suppose, for a contradiction, that s is rational, say $s = q$ for some $q \in \mathbb{Q}$. Since $s > 0$ we have $q \neq 0$, and (3) gives

$$w = \frac{q^3 - 5}{3q} \in \mathbb{Q},$$

contradicting the irrationality of w. Therefore $s = a+b$ is irrational, i.e. $\sqrt[3]{2}+\sqrt[3]{3}$ is irrational. $\square$